

This topic will explain the different project management process groups and their functions. You will explore how project management process groups are related to project management knowledge areas. Next, you will explore a variety of templates for documents required in each project management process group. Finally, you will compare and contrast different methodologies of project management.

You will read the content, watch videos and complete activities to test your understanding of this topic.

But before you begin, let's briefly review the concept of project phase and introduce the project management process groups.

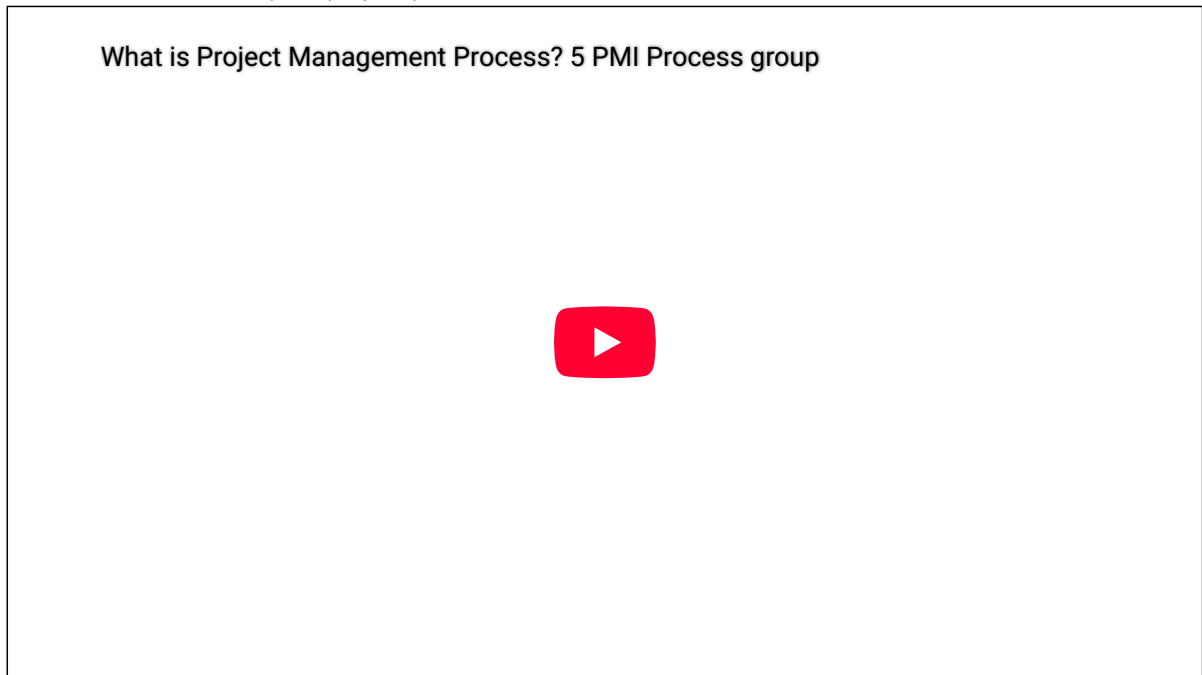
Discuss



What is the project management process?

In this discussion, you will watch a video, answer questions and share your thoughts via the padlet provided.

1. **Watch** the following video "What is the project management process? [5.41 min]," which delves into the five project management processes and reviews the concept of project phase.



© 2021 by Project Management. *What is project management process?*

2. **Answer** the following questions and **share** your thoughts via the padlet below.
 - What are some of the key takeaways from this video?
 - What are the differences between project phase and the project management process?
3. **Review** your classmates' posts. Is there anything in their experience that is new to you? Comment constructively on at least two posts.

o use the padlet

Double click anywhere on the board or tap/click on the pink + to add to the padlet. If you want to open the padlet fullscreen, click on the 'open in new' icon in the top right of the padlet screen. Get creative with photos, videos, LinkedIn profile URLs and more.

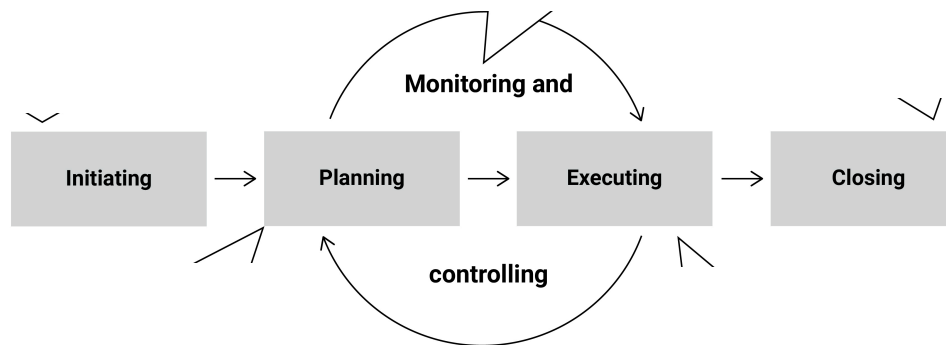
Project management is the process of applying knowledge, skills, resources and technologies to achieve project goals within the identified constraints. Projects are often complex and subject to changes in a highly volatile environment; thus, we need to break down a project into a series of procedural elements to explicitly structure the project and allocate resources with a clear focus. The knowledge, skills, resources and technologies are usually grouped into processes and interact with project phases. PMI's *PMBOK® Guide* (6th edition) identifies five process groups: initiation, planning, execution, monitoring and controlling, and closing (PMI, 2017). These process groups help the project team to plan, manage and execute more effectively with well-defined goals, deliverables, workflows and accountabilities.

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The project lifecycle is the whole life of the project, from start-up/initiation, through to delivery and closure. This can be split down into smaller parts. Project management processes are the processes/actions you go through in order to manage and deliver a project and get it to progress through the project lifecycle.

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(Meer, 2016)



Formally closes the project with acceptance from the customer or sponsor

✓ Check

← References

Jump to...

Project management knowledge areas →

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The main activities of the five process groups can be mapped into 10 project management knowledge areas.

Figure 2.1 showcases the 10 project management knowledge areas.

Figure 2.1
Project management knowledge areas



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In Table 2.1 below, you can also see the relationships between process groups and the knowledge areas.

Table 2.1
Relationship between project management process groups and knowledge areas

Knowledge Areas	Project Management Process Groups				
	Initiating Process Group	Planning Process Group	Executing Process Group	Monitoring and Controlling Process Group	Closing Process Group
4. Project Integration Management	4.1 Develop Project Charter	4.2 Develop Project Management Plan	4.3 Direct and Manage Project Work 4.4 Manage Project Knowledge	4.5 Monitor and Control Project Work 4.6 Perform Integrated Change Control	4.7 Close Project or Phase

Knowledge Areas	Project Management Process Groups				
	Initiating Process Group	Planning Process Group	Executing Process Group	Monitoring and Controlling Process Group	Closing Process Group
5. Project Scope Management		5.1 Plan Scope Management 5.2 Collect Requirements 5.3 Define Scope 5.4 Create WBS		5.5 Validate Scope 5.6 Control Scope	
6. Project Schedule Management		6.1 Plan Schedule Management 6.2 Define Activities 6.3 Sequence Activities 6.4 Estimate Activity Durations 6.5 Develop Schedule		6.6 Control Schedule	
7. Project Cost Management		7.1 Plan Cost Management 7.2 Estimate costs 7.3 Determine Budget		7.4 Control Costs	
8. Project Quality Management		8.1 Plan Quality Management	8.2 Manage Quality	8.3 Control Quality	
9. Project Resource Management		9.1 Plan Resource Management 9.2 Estimate Activity Resources	9.3 Acquire Resources 9.4 Develop Team 9.5 Manage Team	9.6 Control Resources	
10. Project Communications Management		10.1 Plan Communications Management	10.2 Manage Communications	10.3 Monitor Communications	



Knowledge Areas	Project Management Process Groups				
	Initiating Process Group	Planning Process Group	Executing Process Group	Monitoring and Controlling Process Group	Closing Process Group
11. Project Risk Management		11.1 Plan Risk Management 11.2 Identify Risks 11.3 Perform Qualitative Risk Analysis 11.4 Perform Quantitative Risk Analysis 11.5 Plan Risk Responses	11.6 Implement Risk Responses	11.7 Monitor Risks	
12. Project Procurement Management		12.1 Plan Procurement Management	12.2 Conduct Procurements	12.3 Control Procurements	
13. Project Stakeholder Management	13.1 Identify Stakeholders	13.2 Plan Stakeholder Engagement	13.3 Manage Stakeholder Engagement	13.4 Monitor Stakeholder	

© PMI, n.d.

Table 2.2 shows how PMI (Project Management Institute) divided the field of project management into 10 specific knowledge areas necessary for effective project management that overlap with the process groups. We can think of the knowledge areas as vertical, while the process groups are horizontal.

Table 2.2
The 10 project management knowledge areas (PMBOK®)

Integration management	What holds a project together? That would be project integration management, which includes such fundamental plans as developing a project charter that is created during the initiation phase. This is the document that sets up the project and assigns the project manager. Another aspect of this area is the project management plan, which is developed as a project road map for the project to reach a successful end. Once created, the project plan is approved by stakeholders and/or sponsors, and then it's monitored and tracked through a change log as the project progresses. The project integration area also includes the directing and managing of the project work, which is the production of its deliverables. This process is monitored, analysed and reported on to identify and control any changes or problems that might occur. Also, any change control will be carried out. That might require request forms, approval from stakeholders and/or sponsors or another admin. This area is also part of the project closure at the end of the project.
Scope management	Scope relates to the work of the project. So, that includes plan scope management, which is part of the project management plan. It also is when a detailed requirement for the final product or service is collected. You'll also need to define scope in a scope statement. This is anything from a sentence to a bulleted list that is comprehensive to reduce major project risks. And a work breakdown structure (WBS), which is a graphic breakdown of project work, is another part of this area. Validate scope during the project, which means making sure that the deliverables are being approved regularly by the sponsor or stakeholder. This occurs during the monitoring and controlling process groups and is about accepting the deliverables, not the specs laid out during planning. The scope statement is likely going to change over the course of the project to control the scope, such as if a project falls behind schedule.



Time management

Project time management is, no surprise, time consuming. The project is divided into tasks, which are scheduled with start dates and deadlines, as well as budgets for each task. And things are constantly changing over the phases of any project, which means revising these things often.

This involves plan schedule management, which involves creating a schedule for the project and determining who is responsible for what. That means defining activities, which is not the same as making a WBS, but similar. So, you create a task list that touches on every aspect of the project.

These tasks are then put in an order that makes sense, and any dependencies between them is noted. These dependencies are then determined to be either finish-to-start (FS), finish-to-finish (FF), start-to-start (SS) or start-to-finish (SF). This is mostly for larger projects.

With the tasks now sequenced, the resources required for each must be estimated and assigned. The duration of each task is also determined at this point. All this will lead to a schedule by first figuring out the critical path and float for each task. Use a Gantt chart to place the tasks on a timeline, and then work on resource levelling to balance resource usage.

Once the schedule is made, plans to control the schedule are necessary. Earned value management is performed regularly to make sure that the actual plan is proceeding as it had been planned.



Cost management	This area involves the project budget, which means having good estimating tools to make sure that the funds cover the extent of the project and are being monitored regularly to keep stakeholders or sponsors informed. Plan cost management will determine the method to establish the budget, which includes how and if it will change and what procedures will be used to control it. Each task will have to be estimated for cost, which means including all resources such as labour, materials, equipment and anything else needed to complete the task. This will determine the project budget once you take all the task costs and combine them. Then comes the need to control those costs through an earned value analysis. This is performed regularly throughout the project to make sure the estimated costs are in line with actual expenditures.
Quality management	A project can come in on time and within budget, but if the quality is not up to the standard set, then the project is a failure. Plan quality management is part of the overall project management plan, though it can be a standalone document if it contains the quality specs for the product or service. The process needs to include quality assurance, which is just a way to make sure that quality standards are being met. Therefore, to control quality, the deliverables must be inspected to make sure that those standards outlined in the quality management plan are being met.



Human resource management	The project team is your most important resource, so it's crucial to assemble the best team and to make sure they're happy. But also, you need to track their performance to ensure that the project is progressing as planned. A human resource management plan will identify their roles and their requirements for those positions, as well as how they fit in the overall project structure. After you've determined the job descriptions, it's time to fill those positions and acquire a project team. This can be done in-house by drawing from other departments in the organisation, by getting new hires or by a combination of both. The team needs development, possibly training and other things that will make them viable for the project. Managing the project team is an ongoing responsibility of the project manager. The team is monitored to make sure they're working productively and that there are no internal conflicts, so everyone is satisfied.
Communications management	All areas of project management are important, but communication management might be paramount as it informs every aspect of the project. Communications inform the team and stakeholders, therefore the need to plan communications management is a critical step in any project. It is at this point that the dissemination of communications is determined, including how it's done and with what frequency. Target who needs what and when. Also, note how communications will occur when issues arise in the project, such as changes. Manage the communications when the project is executed to make sure it runs as planned. This will also involve controlling communications by reviewing their effectiveness regularly and adjusting as needed.



Risk management	Risk management plans will identify how the risks will be itemised, categorised and prioritised. This involves identifying risks that might occur during the execution of the project by making a risk register. Perform qualitative risk analysis after the biggest risks have been identified and classified by likelihood and impact. Then prioritise them. Then perform quantitative analysis according to their impact on the project, such as its budget, schedule and so on. Now you'll need to plan risk responses. If those risks in fact become issues, then a response needs to have been written in advance, with an owner who can make sure the risk is properly identified and handled. Controlling risk involves regularly reviewing the risk register and crossing off those risks that are no longer going to impact the project.
Procurement management	This deals with outside procurement, which is part of most projects, such as hiring subcontractors. This will obviously have an impact on the budget and schedule. Planning procurement management starts by identifying the outside needs of the project and how those contractors will be involved. Now conduct those procurements by hiring the contractors, which includes a statement of work, terms of reference, request for proposals and choosing a vendor. You'll want to control the procurement process by managing and monitoring, and then closing the contracts once the work has been done to everyone's satisfaction.



Stakeholder management	The stakeholders must be happy, as the project has been created for their needs. Therefore, they must be actively managed like any other part of the project. To start one must identify the stakeholders through stakeholder analysis. It's not always easy, but it's a crucial part of starting any project, so find out who they are and what concerns they have. Now plan stakeholder management, which means listing each stakeholder and prioritising what their concerns are and how they might impact the project. This will lead to managing stakeholders' expectations to make sure their needs are met and that you're in communication with them. Throughout the project, control stakeholder engagement. Do this by determining if the stakeholders' needs are being addressed. If not, figure out what changes need to be made to either satisfy those needs or adjust the expectations.
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Adapted from [The 10 Project Management Knowledge Areas](#) © ProjectManager.com, Inc., 2019

 Read

Learn about the role of implementing BIM applications (software technology) in enhancing project management knowledge areas in Egypt's construction industry by reading the article [The role of implementing BIM applications in enhancing project management knowledge areas in Egypt](#) (Shaqour, 2022).

← Topic 3: Project management process groups

Jump to...

Discuss: Why do we need process groups and project management knowledge areas? →

Each project management process group has corresponding documentation required to advance through the project life cycle. Let's look at them:

- **Initiation:** business case, feasibility study report, project charter, kick-off meeting.
- **Planning:** project plan, project schedule, project budget, work breakdown structure (WBS), scope management plan, risk register, resource management plan, stakeholder management plan, performance measurement baseline (PMB).
- **Executing:** task lists, schedule management charts, project costs, resource planning.
- **Monitoring and controlling:** project management reports, risk assessments, KPI dashboards, critical path method (CPM), earned value management.
- **Closing:** project closure report.

Let's explore some practical templates below.

Resources

Explore and download helpful project management templates by visiting [Free project management templates](#) (Project Management Docs, n.d.) and [Free project management templates](#) (ProjectManager, n.d.).

← Discuss: Why do we need process groups and project management knowledge areas?

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IT project management methodologies are a set of practices and principles that assist you during the development of your projects to ensure their best implementation. As a result, organisations use frameworks that better adapt to their structure and culture to deliver a project effectively. Therefore, it is fundamental for them to map a project management methodology correctly from the very beginning of the project.

Different project management methodologies include Waterfall, Agile, Scrum, Kanban, Project Management Body of Knowledge (PMBOK®), Extreme Programming (XP), Adaptive Project Framework (APF), RUP, PRINCE2, Lean, Six Sigma and more.

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The benefits of establishing a project management methodology are:

- *Team members have the same guidelines for their workflow, producing consistent results throughout the project.*
- *Your team is prepared ahead of time to address issues and troubleshoot potential problems instead of simply reacting to emergency situations.*
- *Making adjustments to your workflow is easier when you have an existing methodology to compare changes to.*
- *You can explain your project approach to stakeholders and clients.*
- *Team priorities are always the same, and individuals understand the impact of their contributions.*

”

(Indeed, n.d.)

Watch the video below for an outline of the main project management methodologies, including their advantages and disadvantages.



Watch

Watch this video "Project management methodologies [12.49 min]" to learn about project management methodologies from the traditional to the more flexible framework.

Project Management Methodologies – Learn With Wrike



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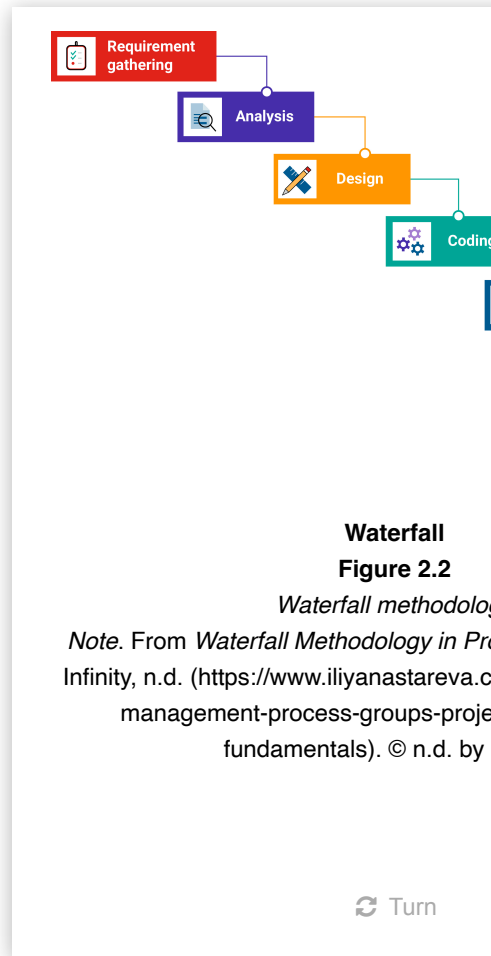
Let's explore the activity below to help you remember important project management methodologies.



Activity



Turn the cards for a brief description of each project management meth



Card 1 of 4

Out of the methodologies mentioned above, Scrum is a very popular Agile approach. It is very important for you to understand Agile as a trendy option of project management methodology as you will use this methodology often in your career as a project manager.

Watch

The video below is an excellent resource for you to understand Scrum and how it can be useful in the project management process. Watch this video "What is Scrum? [1.20 min]" to learn more about Scrum.

What is Scrum?



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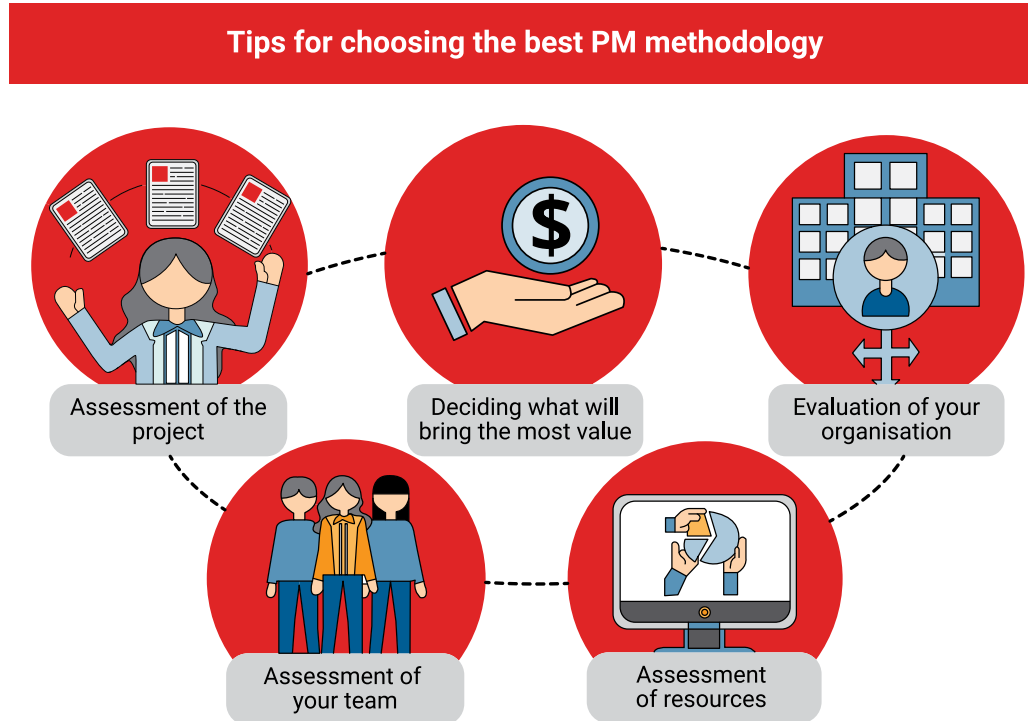


Choosing the right project management methodology

Many aspects play a role when choosing the proper project management methodology. Still, there are some essential factors you will need to address, as shown in Figure 2.6.

Figure 2.6

Basic considerations for choosing the ideal project management methodology



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Good methodologies help integrate project management processes with other processes and cultures of the organisation. These repetitive processes or frameworks are tested project management experiences across industries over the years, which are essential means to achieving project management excellence and maturity.

**Read**

The following articles provide excellent ideas for choosing the right methodologies. Read [Choosing the right path: We asked the project management community: How do you decide which project delivery approach to take?](#) (PM Network, 2018) and [Essential project management methodologies and when to use them](#) (Hart, 2022) to understand how to choose a precise delivery approach for a project.

Out of the methodologies mentioned in the readings above, two of the most commonly used methodologies are Waterfall and Agile. There is often debate between project managers on which one is better. The article below will help you understand both these methodologies and their application in detail.

**Read**

The article [Agile vs. Waterfall: Which project management methodology is best for you?](#) (Hoory & Bottorff, 2022) provides a very clear summary of the two approaches. Reading this article will help you in your role as a project manager, where you are required to make important decisions regarding the use of methodologies.



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